Pricing Intelligence Model – Methodology

Project: Competitive Laptop Pricing Intelligence Model

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1. Data Source

The dataset comprises multi-brand laptop pricing and performance attributes across major market players.

Key variables include:

- **Brand**
- **Product Model**
- **Price (₹)**
- **Estimated Units Sold**
- **Hardware Specifications (Processor, RAM, Storage, Graphics)**
- **Strategic Pricing Classification**

Data was cleaned and standardized prior to modeling.

Data Preparation Steps

- **Null value validation**
- **Outlier inspection for extreme price distortions**
- **Standardization of price format (₹)**
- **Aggregation at brand level for strategic mapping**
- **Validation of revenue totals against dashboard KPIs**

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2. Revenue Modeling Framework

Revenue was calculated using:

Revenue = Estimated Units Sold × Price

Brand-level revenue was aggregated to evaluate market concentration and dominance patterns.

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3. Performance Scoring Model

A weighted composite performance score was developed to objectively measure product capability.

Components:

- Processor Benchmark Score
- RAM Capacity
- Storage Configuration (SSD/HDD weighting)
- Graphics Capability

Each component was weighted proportionally to reflect performance contribution in consumer electronics purchasing decisions.

This generated a normalized Performance Score for each product.

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4. Value Index Calculation

The Value Index was calculated as:

Value Index = Performance Score / Price

This metric quantifies performance delivered per unit price.

Products were segmented into:

- High Value
- Balanced
- Low Value

Segmentation thresholds were derived using percentile-based distribution logic.

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5. Strategic Market Positioning Framework

Brand positioning was determined using a two-dimensional competitive matrix:

- **X-axis → Average Price**
- **Y-axis → Total Revenue**

Median reference lines were applied to classify brands into four quadrants:

Quadrant	Interpretation
Premium Leaders	High Price + High Revenue
Value Volume Leaders	Lower Price + High Revenue
Weak Premium	High Price + Low Revenue
Low Impact	Low Revenue + Low Price

This framework supports structured strategic classification.

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6. Pricing Elasticity Simulation

A dynamic pricing simulator was implemented to evaluate revenue sensitivity under controlled pricing adjustments.

Simulation Logic:

Adjusted Revenue = (Adjusted Price × Adjusted Units)

Unit adjustments were modeled using elasticity assumptions derived from consumer electronics pricing sensitivity patterns.

Scenario applied:

- **±3% pricing adjustment**

Observed Impact:

- **3% price change resulted in ~4.6% revenue variation.**

This indicates measurable demand sensitivity.

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7. Dashboard KPIs

The following key metrics were implemented:

- **Total Revenue**
- **Average Selling Price**
- **High Value Share (%)**
- **Average Value Index**
- **Market Share (%)**
- **Simulated Revenue**
- **Revenue Impact (%)**

These KPIs were used to support executive-level strategic interpretation.

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8. Limitations

- **Units sold are estimated and not derived from proprietary sales data.**
- **The elasticity factor is modeled assumption-based.**
- **Market dynamics such as seasonality and channel distribution were not incorporated.**

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9. Strategic Application

The Pricing Intelligence Model enables:

- **Identification of revenue concentration risk**
- **Detection of underpriced high-value opportunities**
- **Structured premium pricing defense**
- **Data-backed scenario simulation**

This framework is designed to support pricing optimization, competitive positioning, and revenue growth planning in consumer electronics markets.